

Bangalore Institute of Technology

K. R. Road, VV Pura, Bangalore 560004

Department of Information Science and Engineering

Parallel Computing Laboratory (BCS702)

PREREQUISITE:

1. Knowledge of C programming
2. Operating System
3. Data Structures and Algorithms
4. Computer Organization and Architecture

COURSE OUTCOMES:

VII-SEM Course Name: Parallel Computing Course Code: BCS702 Year of Study: 2026-27	
At the end of the course the students should be able to:	
CO1	Understand parallel programming concepts like classifications of parallel computers, various architecture, memory models and tools.
CO2	Apply parallel programming paradigms using MPI, OpenMP, and CUDA to solve computational problems.
CO3	Analyze the performance of parallel programs by evaluating various performance metrics.
CO4	Implement and demonstrate parallel programming constructs using OpenMP/MPI/CUDA for shared and distributed memory systems.
CO5	Design and implement solutions for real time problems by integrating parallel programming techniques.

Mapping of COs-POs and COs-PSOs

Parallel Computing Laboratory (BCS702)

CO's with PO's and PSO's Mapping														
VII-SEM		Course Name: Parallel Computing Course Code: BCS702 Year of Study: 2026-27												
Cos	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2													
CO2	3												2	2
CO3	3	2											2	2
CO4	3	2			3								2	2
CO5	3	2	2		3				3	3			2	2
AVG	2.8	2	2		3				3	3			2	2

Parallel Computing Laboratory

List of Programs

1. Write a OpenMP program to sort an array on n elements using both sequential and parallel mergesort(using Section). Record the difference in execution time.
2. Write an OpenMP program that divides the Iterations into chunks containing 2 iterations, respectively (OMP_SCHEDULE=static,2). Its input should be the number of iterations, and its output should be which iterations of a parallelized for loop are executed by which thread.

For example, if there are two threads and four iterations, the output might be the following:

- a. Thread 0 : Iterations 0 — 1
 - b. Thread 1 : Iterations 2 — 3
3. Write a OpenMP program to calculate n Fibonacci numbers using tasks.
 4. Write a OpenMP program to find the prime numbers from 1 to n employing parallel for directive. Record both serial and parallel execution times.
 5. Write a MPI Program to demonstration of MPI_Send and MPI_Recv.
 6. Write a MPI program to demonstration of deadlock using point to point communication and avoidance of deadlock by altering the call sequence.
 7. Write a MPI Program to demonstration of Broadcast operation.
 8. Write a MPI Program demonstration of MPI_Scatter and MPI_Gather.
 9. Write a MPI Program to demonstration of MPI_Reduce and MPI_Allreduce (MPI_MAX, MPI_MIN, MPI_SUM, MPI_PROD)

Parallel Computing Laboratory Evaluation Rubrics

Subject Code: BCS702

CIE Marks: 25

Number of Contact Hours/Week: 2

Lab Write-up and Execution rubrics for Daily Conduction (Max: 15 Marks)

Sl. No	Task	Good	Average
a.	Understanding of problem and approach to solve (4 Marks)	Demonstrate good knowledge of design concepts of given problem and implementation. (4)	Moderate understanding of the algorithm and its implementation.(2)
b.	Execution (5 Marks)	Program has all possible conditions while execution with proper results. (5)	Average conditions are defined and verified during execution. (3)
c.	Analysis (3 Marks)	Investigate the various Algorithmic technique.(3)	Investigate a couple of Algorithmic technique.(2)
d.	Record (3 Marks)	Lab Record submitted as directed and on time (3)	Directions were not followed or submitted on time (2)

Lab Write-up and Execution rubrics for Internals (Max: 25 Marks)

Sl. No	Task	Good	Average
a.	Understanding of problem and approach to solve (6 Marks)	Demonstrate good knowledge of design concepts of given problem and implementation. (6)	Moderate understanding of the algorithm and its implementation in Java. (3)
b.	Execution (10 Marks)	Program has all possible conditions while execution with proper results. (10)	Average conditions are defined and verified during execution. (5)
c.	Analysis (4 Marks)	Investigate the various Algorithmic technique (4)	Investigate a couple of Algorithmic technique. (2)
d.	Viva (5 Marks)	Explains different steps of algorithm/problem and related concepts involved.(5)	Adequately provides explanation on algorithm/problem.(3)

Parallel Computing Laboratory

Subject Code: BCS702

CIE Marks: 25

Number of Contact Hours/Week:2

Lesson Planning / Schedule of Experiments

Sl. No	Name of the Experiment	To be completed
1	Sample Programs	Week 1
2	1. Write a OpenMP program to sort an array on n elements using both sequential and parallel mergesort(using Section). Record the difference in execution time.	Week 2
3	2. Write an OpenMP program that divides the Iterations into chunks containing 2 iterations, respectively (OMP_SCHEDULE=static,2). Its input should be the number of iterations, and its output should be which iterations of a parallelized for loop are executed by which thread. For example, if there are two threads and four iterations, the output might be the following: a. Thread 0 : Iterations 0 --- 1 b. Thread 1 : Iterations 2 --- 3	Week 3
4	3. Write a OpenMP program to calculate n Fibonacci numbers using tasks.	Week 4
5	4. Write a OpenMP program to find the prime numbers from 1 to n plying parallel for directive. Record both serial and parallel execution times.	Week 5
6	Test-I	Week 6
7	5. Write a MPI Program to demonstration of MPI_Send and MPI_Recv.	Week 7
8	6. Write a MPI program to demonstration of deadlock using point to point communication and avoidance of deadlock by altering the call sequence.	Week 8
9	7. Write a MPI Program to demonstration of Broadcast operation.	Week 9
10	8. Write a MPI Program demonstration of MPI_Scatter and MPI_Gather.	Week 10
11	9. Write a MPI Program to demonstration of MPI_Reduce and MPI_Allreduce (MPI_MAX, MPI_MIN, MPI_SUM, MPI_PROD)	Week 11
12	Test-II	Week 12

Sl No	Contents	Page No.	Date of Execution	Date of Submission on	Daily Conduction				Total Marks	Staff Signature
					Understanding 4M	Execution 5M	Analysis 3M	Record 3M		
1	Merge Sort									
2	OpenMP program that divides the Iterations into chunks									
3	Calculate n Fibonacci numbers									
4	Prime numbers									
5	MPI Program to demonstration of MPI_Send and MPI_Recv									
6	Demonstration of deadlock									
7	Demonstration of Broadcast operation									
8	Demonstration of MPI_Scatter and MPI_Gather									
9	Demonstration of PI_Reduce and MPI_Allreduce (MPI_MAX, MPI_MIN, MPI_SUM, MPI_PROD)									
Total										
Average										
Average of Daily Conduction										

LAB INTERNAL MARKS				
TEST - 1 (25 Marks)	Write up 10M	Execution 10M	Viva 5M	Total Marks
TEST - 2 (25 Marks)	Write up 10M	Execution 10M	Viva 5M	Total Marks
Average Marks				

FINAL LAB MARKS		
	Max marks	Marks scored
Average of Daily Conduction	15	
Test average marks	10	
TOTAL	25	
Signature of the Faculty		